

ENERGY FLOW IN AN ECOSYSTEM

The transfer of energy from the source in plants through a series of organisms by eating and being eaten constitutes **food chains**. At each transfer, a large proportion of energy is lost in the form of heat. These food chains are not isolated sequences but are inter-connected with each other. This interlocking pattern is known as the **food web**. Each step of the food web is called a **trophic level**. Hence green plants occupy the first level, herbivores the second level, carnivores the third level and secondary carnivores the fourth level. These trophic levels together form the ecological pyramid.

The food chains

The most obvious aspect of nature is that energy must pass from one living organism to another. When herbivorous animals feed on plants, energy is transferred from plants to animals. In an ecosystem, some of the animals feed on other living organisms, while some feed on dead organic matter. At each linkage in the chain, a major part of the energy from the food is lost for daily activities. Each chain usually has only four to five such links. However, a single species may be linked to a large number of species.

The food chains are mostly sequential and begin from green plants followed by herbivores and two successive sets of carnivores or predate. Such food chains are called Grazing food chain or Predator food chain (Fig. 1a). In addition, the food chains that start from dead organic materials that are consumed by a series of parasites and hyperparasites is called Detritus food chain or the Saprophytic food chain (Fig. 1b). These food chains are invariably linked to one another in nature (Fig. 1c).

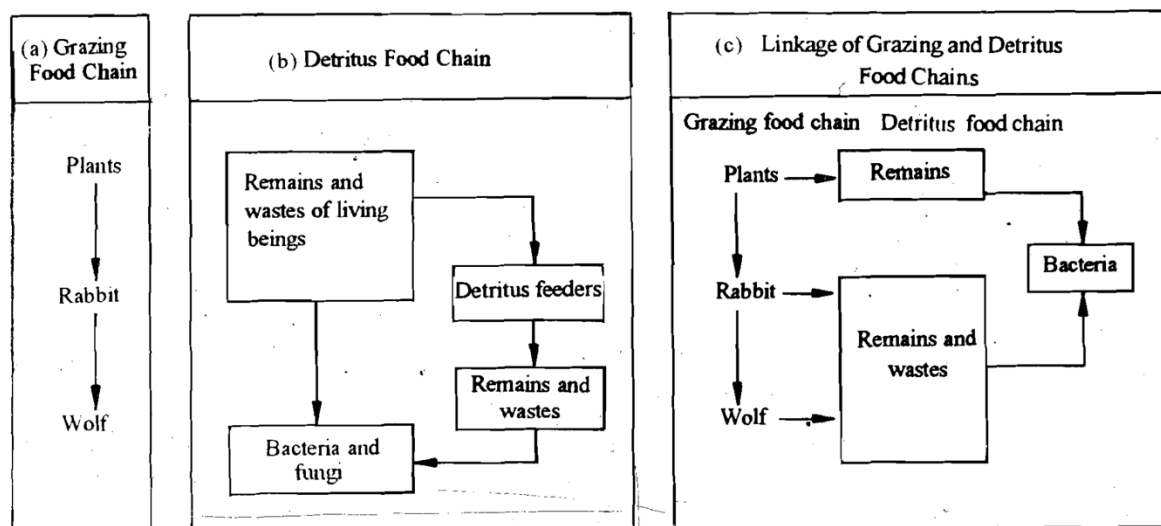


Figure 1. (a) Grazing food chain; (b) Detritus food chain. Bacteria and other organisms fed on plants and animals remain; (c) The Grazing and Detritus food chains are linked.

How Food Chains Work

Every biological community can have multiple and diverse food chains, but every food chain starts with a primary source of energy. The most obvious source of energy is the big ball in the sky, the sun. Other food chains may begin with a boiling-hot deep sea vent as a source of energy.

The next organism to benefit from this initial source is called the primary producer. These are organisms that can create their own food from the main energy source. Some examples include plants and algae. For example, plants are primary producers because they can harness and use the energy from the sun through a process called photosynthesis.

After the plant goes through the work of photosynthesis, another organism may come along and eat the plant, taking its energy to use as its own. As human beings, we are not primary producers because we cannot create our energy to survive, and must consume energy from other sources, like plants. By eating plants, we are part of the next sequence in the food chain, called the primary consumer, or organisms that consume primary producers.

With each transition of energy, the food chain moves up levels. These levels are called trophic levels. Here is a list of the order of trophic levels.

Primary Producers: The one that gathers energy from an energy spot such as the sun; an example may be grass.

Primary Consumer: The one that gets its energy directly from the primary producer, such as a grasshopper who eats the grass

Secondary Consumer: The one that gets its energy directly from the primary consumer, such as the rat who eats the grasshopper

Tertiary Consumer: The one that gets its energy directly from the secondary consumer, such as the snake who eats the rat

Quaternary Consumer: I think you are catching on now. This is the one that gets its energy directly from the tertiary consumer, such as the hawk that eats the snake.

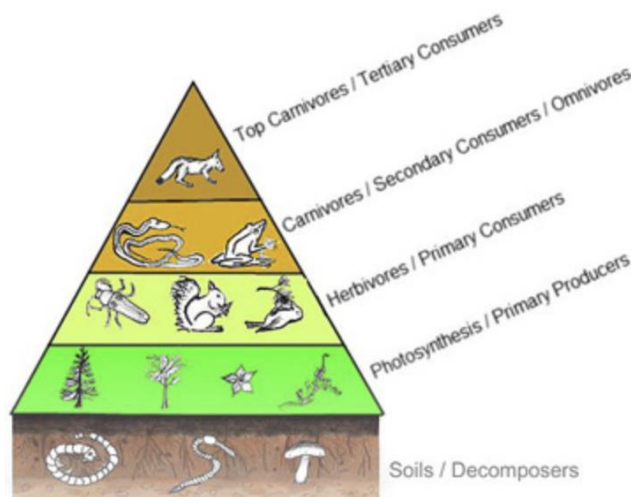


Figure 2. Examples of trophic levels some species may be on.

The Flow of Energy in the Food Chain

As we go along the trophic levels, at each step, a large portion of the energy is lost as heat and only a small fraction about 10% goes on to the next level. Therefore, the quantity of energy decreases successively at a rapid rate from primary producers to the top consumer (carnivores). This explains why the food chains have fewer links. After the fourth or fifth link, not enough energy is available to support another trophic level.

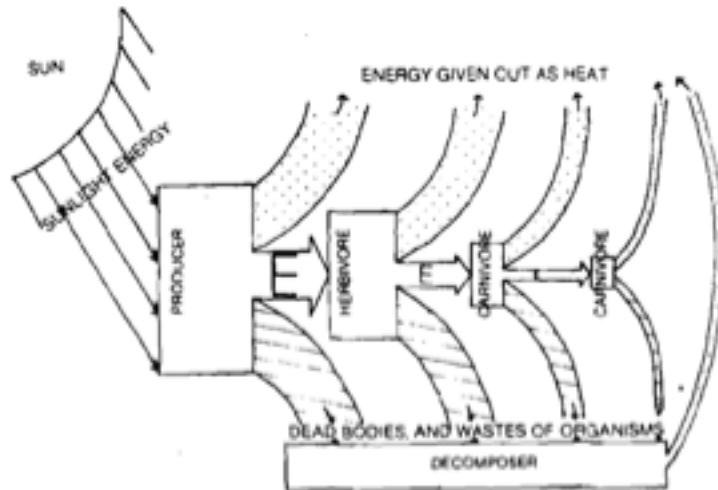


Figure 4. Energy flow in a food chain.

Extending this concept a bit further, in case we want to support more human population on earth, this could be possible by providing them with a vegetarian diet rather than non-vegetarian food. This way maximum energy can be made available as it involves one step in the energy transfer from a primary producer, thus minimizing energy loss at subsequent transfer.

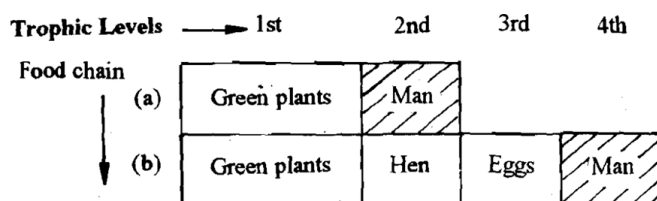
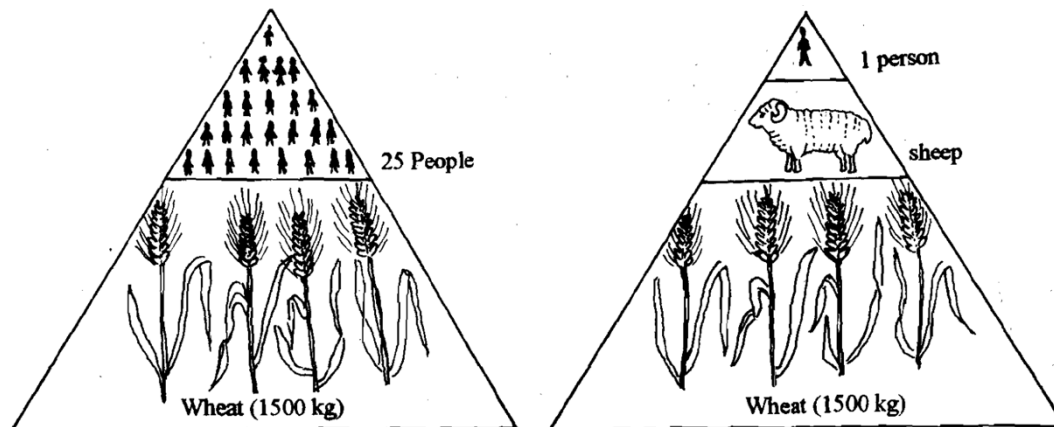


Figure 4. Graphic Illustration of vegetarian and carnivorous diet for supporting the human population.

The Food Web

From the above discussion, you should not be tempted to believe that the food relations in an ecosystem are simple having only linear food chains. Actually, in nearly all natural ecosystems, the patterns of consumption are so complicated that there are many cross-links connecting various organisms. So when the consumers have more than one food source, this results in branching off of food chains. In this way chains become interconnected to form a food web (Fig. 5). The food web is a composite of all the food chains giving us a complete picture of who consumes whom own ecosystem. Food webs represent the transfer of energy and nutrients among the organisms through the ecosystem, whereas a food chain traces only one pathway or represents one strand of the food web.

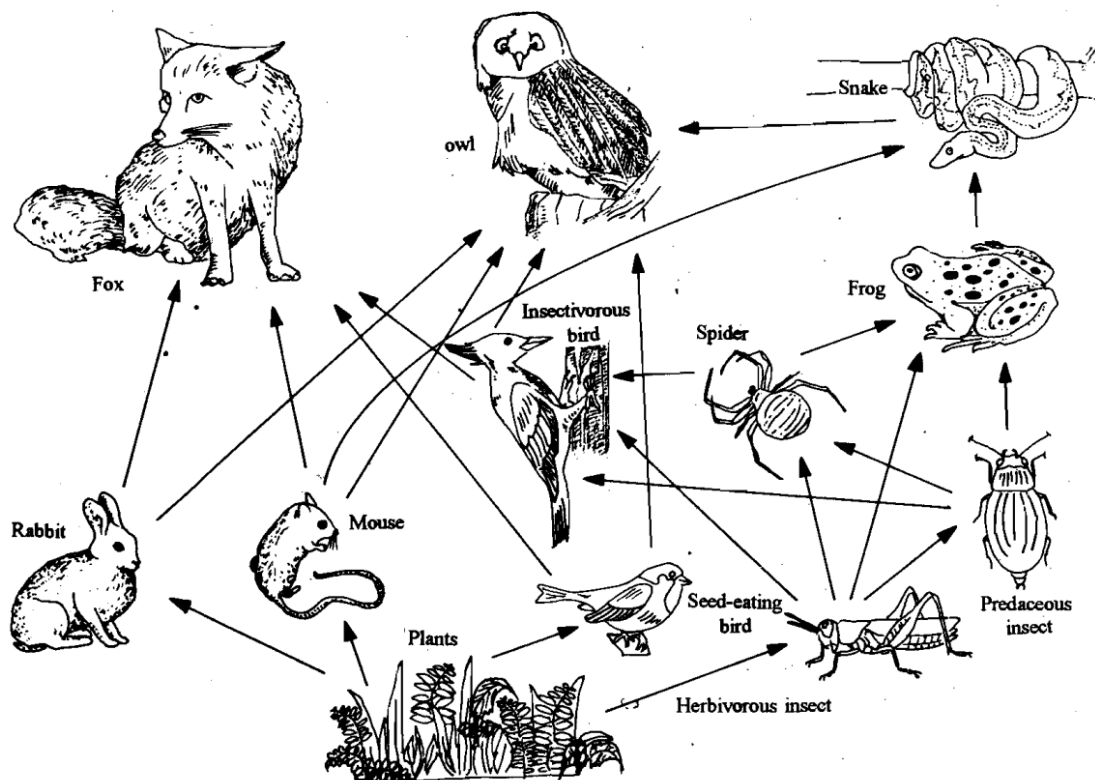


Figure 5. A simplified food web showing the interconnected network of food chains.

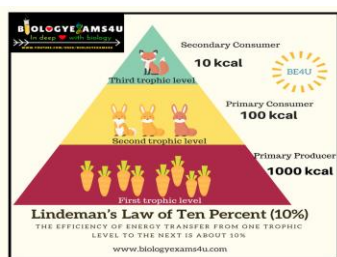
In a food web, many species can occupy more than one trophic level. They are known as multilevel consumers or omnivores. Humans are good examples of this situation. Humans, when they consume food derived from plants - they are primary consumers occupying the 2nd trophic level, and when they consume animal products they may occupy 3rd or higher trophic levels.

Energy flow in the ecosystem

- energy is needed for each and every biological activity
- The transfer of energy from one trophic level to another trophic level is called energy flow. Solar energy is transformed into chemical energy by process called photosynthesis. In a biological world the energy flows from the sun to the plants and

then to all heterotrophic organisms like Nitro- organisms, that is energy flows from the producers to the consumers

- Only 1% of total light falling on the green plants is utilized for photosynthesis. This is sufficient to maintain all life on this earth. There is no 100% flow of energy from one trophic level to the other. Some energy is always lost to the environment, because of this energy cannot be recycled in an ecosystem. It can only flow one way and can never take place in the reverse direction. Energy flow is unidirectional.
- Sun is the ultimate source of energy. A large amount of energy is lost at each Trophic level. It is estimated that upto 90% energy is lost during transfer from one trophic level to another. Therefore the amount of energy available decreases from each trophic level to the other. When a food chain is very small or short the final consumer may receive a large amount of energy, but in a food chain which is long the final consumer may get very less energy.
- This law of 10% was proposed by Lindeman in the year 1942 says that only 10% of food energy is transferred to the next level of consumers the rest is wasted.



Source courtesy:biologyexams4u.com

The illustration shows the progressive loss of energy in a food chain

The flow of energy follows the **two** laws of thermodynamics

The **I law** of thermodynamics states that energy can neither be created nor be destroyed but can be transformed from one form to another

For example: the plants which are the producers utilize the solar energy and convert this energy through photosynthesis into biochemical energy, later the consumers feed on the plants and uses this biochemical energy for their mechanical activities

II law of the second law of thermodynamics: states that energy transformation involves degradation or dissipation of energy from a concentrated to a dissipated form .We see that energy is lost at each and every trophic level

This energy flow supplies energy to all organisms at each trophic level.

Consequences of Food Webs: Biological Magnification

One of the most important consequences of ecosystem dynamics in terms of human impact is biomagnification. **Biomagnification** is the increasing concentration of persistent, toxic substances in organisms at each successive trophic level. These are substances that are fat soluble, not water soluble, and are stored in the fat reserves of each organism. Many substances have been shown to biomagnify, including classical studies with the pesticide dichlorodiphenyltrichloroethane (DDT), which was described in the 1960s bestseller, **Silent Spring by Rachel Carson**. DDT was a commonly used pesticide before its dangers to apex consumers, such as the bald eagle, became known. In aquatic ecosystems, organisms from each trophic level consumed many organisms in the lower level, which caused DDT to increase in birds (apex consumers) that ate fish. Thus, the bird's accumulated sufficient amounts of DDT to cause fragility in their egg shells. This effect increased egg breakage during nesting and was shown to have devastating effects on these bird populations. The use of DDT was banned in the United States in the 1970s.

Other substances that biomagnify are polychlorinated biphenyls (PCB), which were used as coolant liquids in the United States until their use was banned in 1979, and heavy metals, such as mercury, lead, and cadmium. These substances are best studied in aquatic ecosystems, where predatory fish species accumulate very high concentrations of toxic substances that are at quite low concentrations in the environment and in producers.